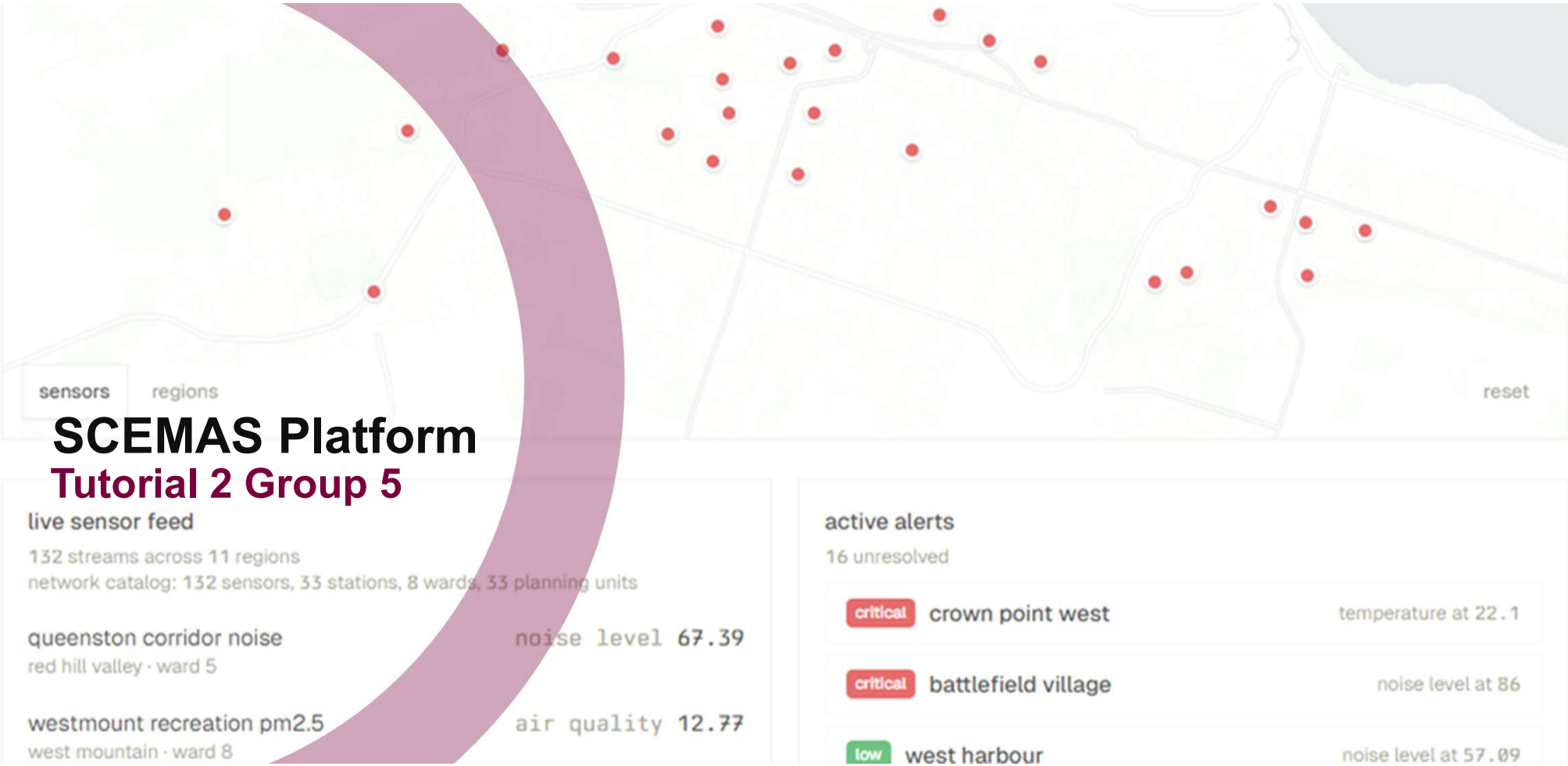

SCEMAS Platform

Tutorial 2 Group 5

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April 7th, 2026

Brighter World

McMaster University

Key Problems our Platform Addresses

Problem	Consequence
Disconnected, manual monitoring	Delayed response to environmental hazards
No real-time threshold detection	Operators inspect raw data manually
No public data access	Citizens unaware of hazards

Viewpoints

- City Operator
- System Admin
- Public
- Third-Party Developer
- Security

Key Requirements

Functional Requirements

ID	Business Event	Who Benefits?
BE1	Signup and Login	All users
BE2	Request and Interpret Environmental Data	Public users & third-party developers
BE3	Define Alert Rules	System administrators
BE4	Acknowledge a Critical Environmental Alert	City operators
BE5	Manage Personalized Alert Subscriptions	City operators

BE5 corresponds to our Innovative Feature:

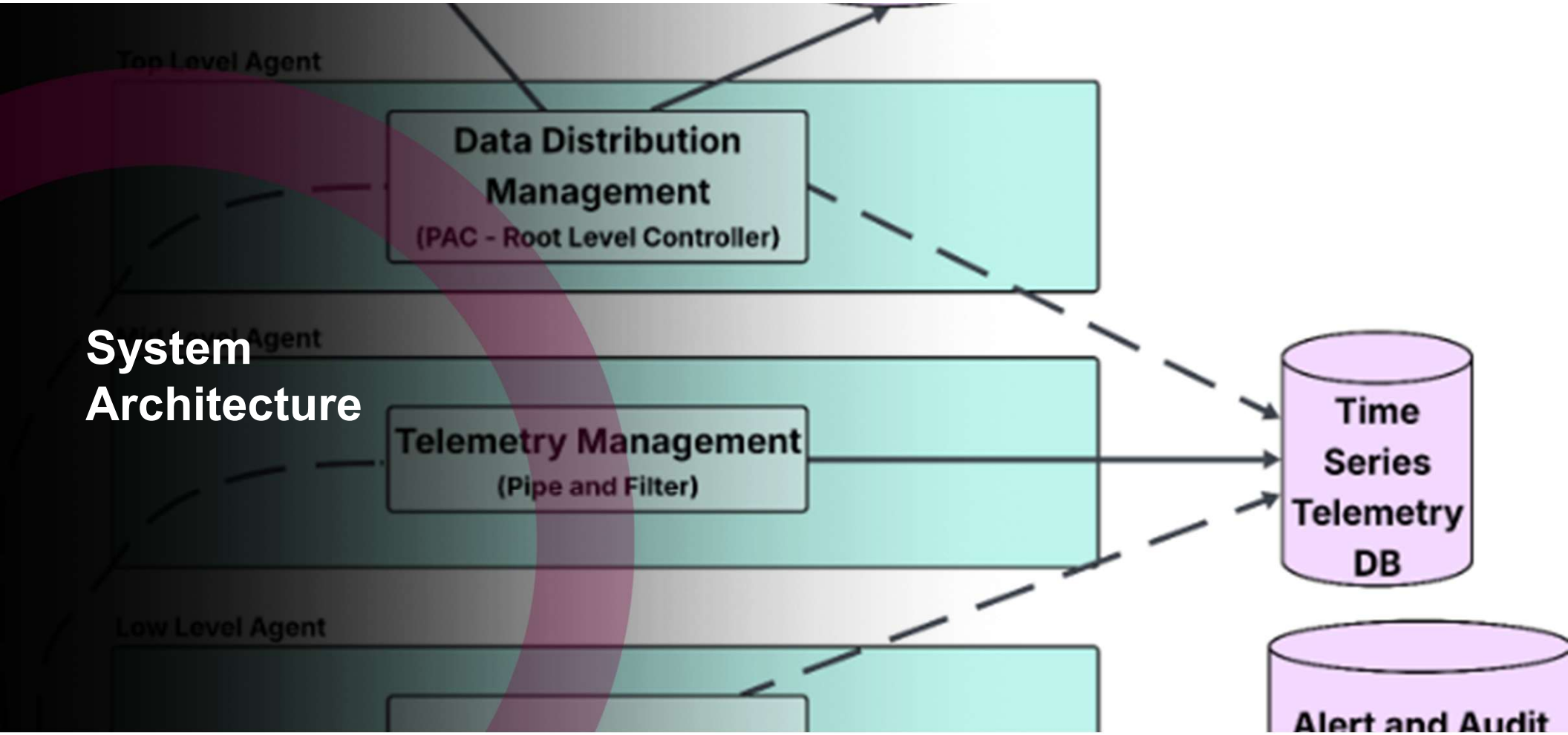
City Operators can manage personalized alert subscriptions

Key Requirements

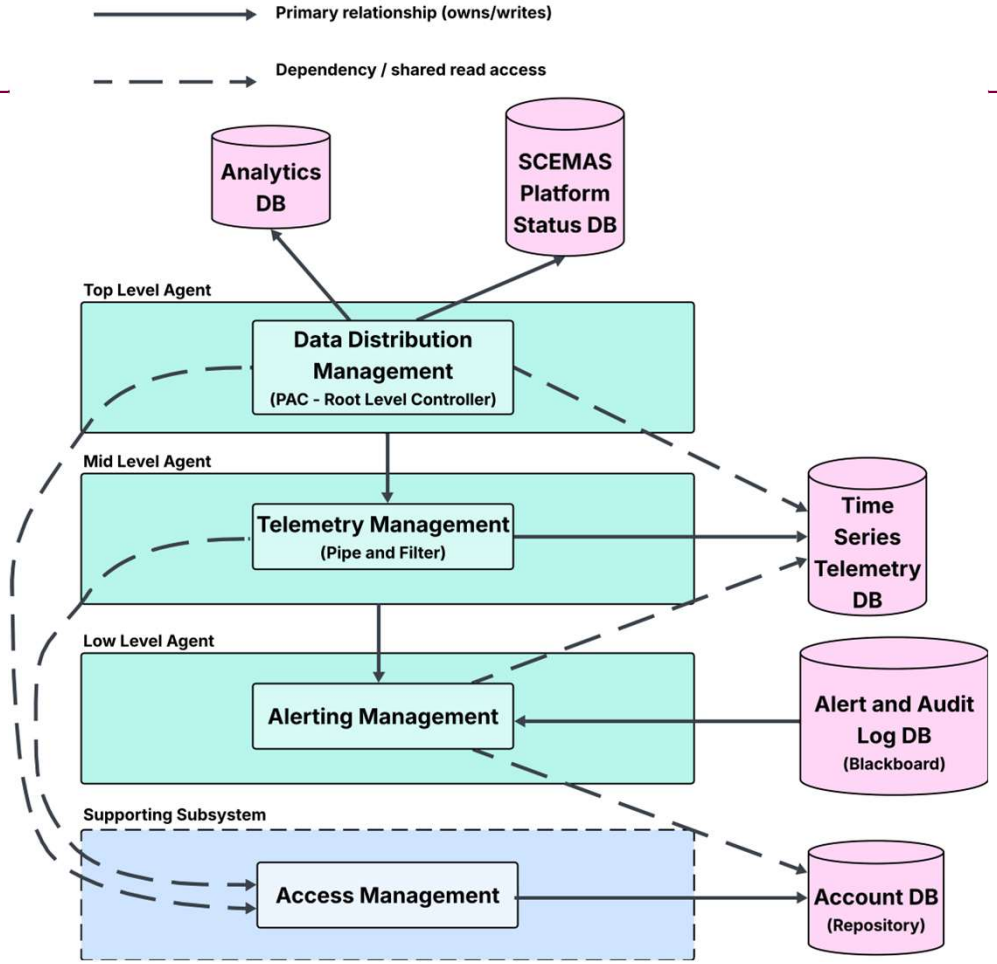
Non-Functional Requirements (NFR)

NFR Class	NFR	Value	Architectural Impact
Speed and latency	Alert generation latency	≤ 5 sec	Blackboard: parallel rule evaluation
Speed and latency	Public API response time	≤ 500ms (p95)	Public REST API layer
Usability	Alert acknowledgement time	≤ 30 sec	PAC: isolated operator view
Privacy	No Personally Identifiable Information collected	-	Data aggregated by zone only

System Architecture

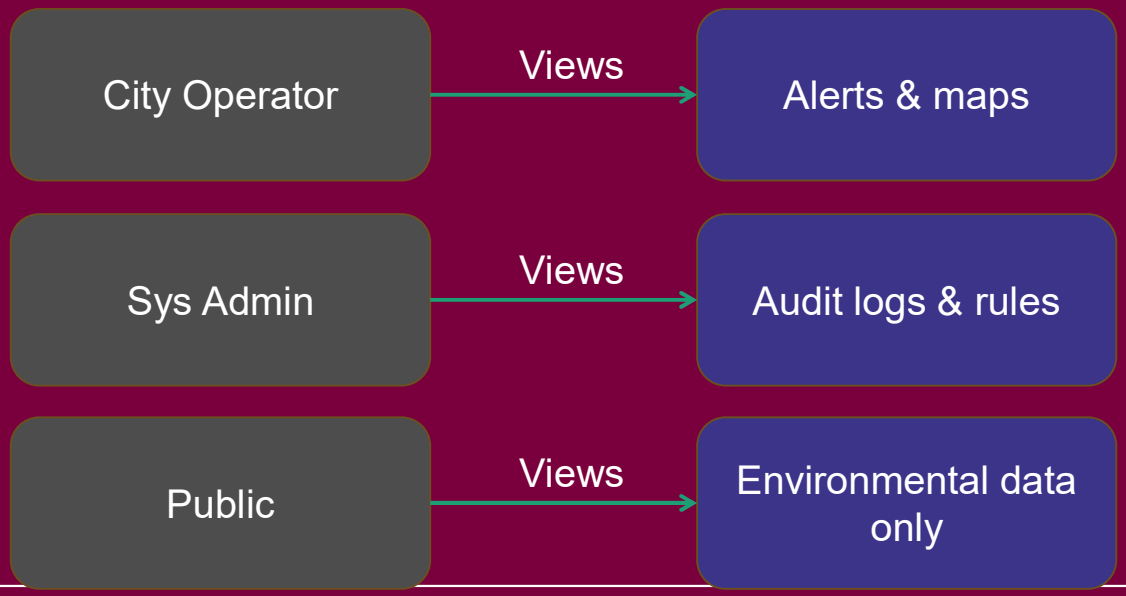


System Architecture Diagram



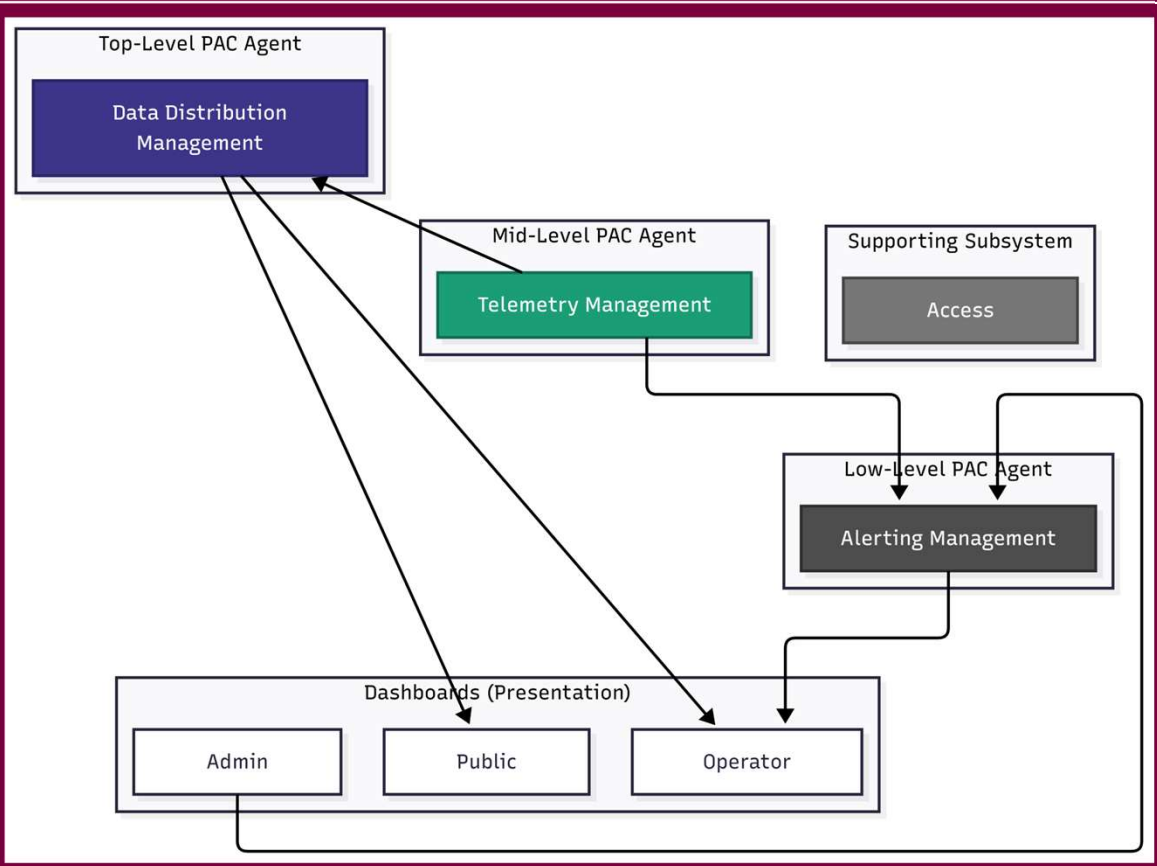
Rationale for Architecture Choices

What?	Architecture Style	Why?
Overall	PAC	• Operator, Admin, Public each needs isolated views • Each agent owns its own data • PAC prevents shared model coupling



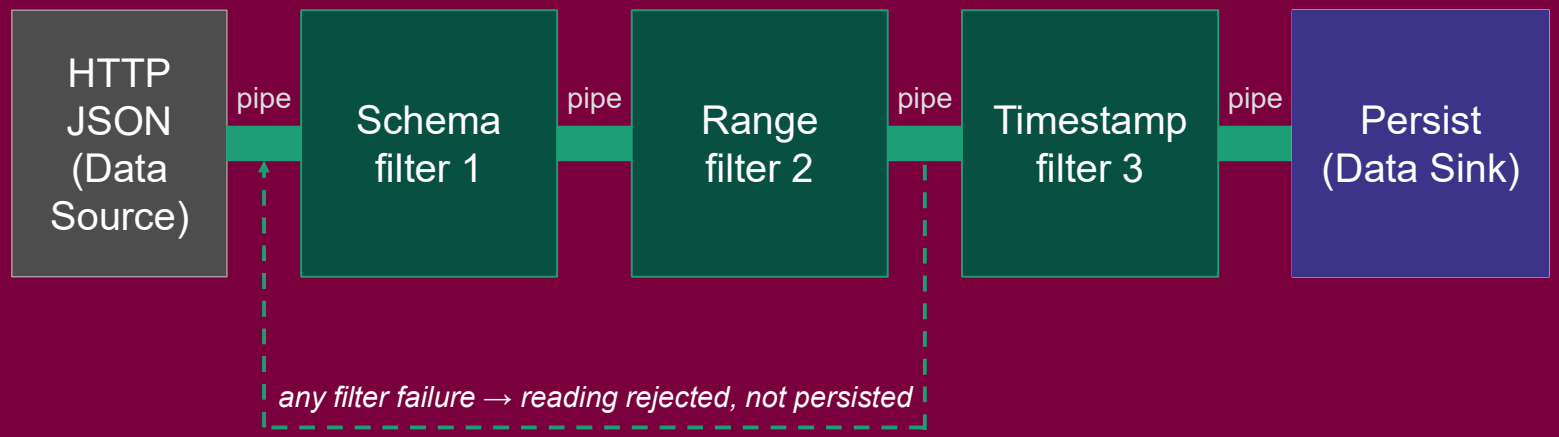
Rationale for Architecture Choices

Mapping PAC Agents to User Dashboards



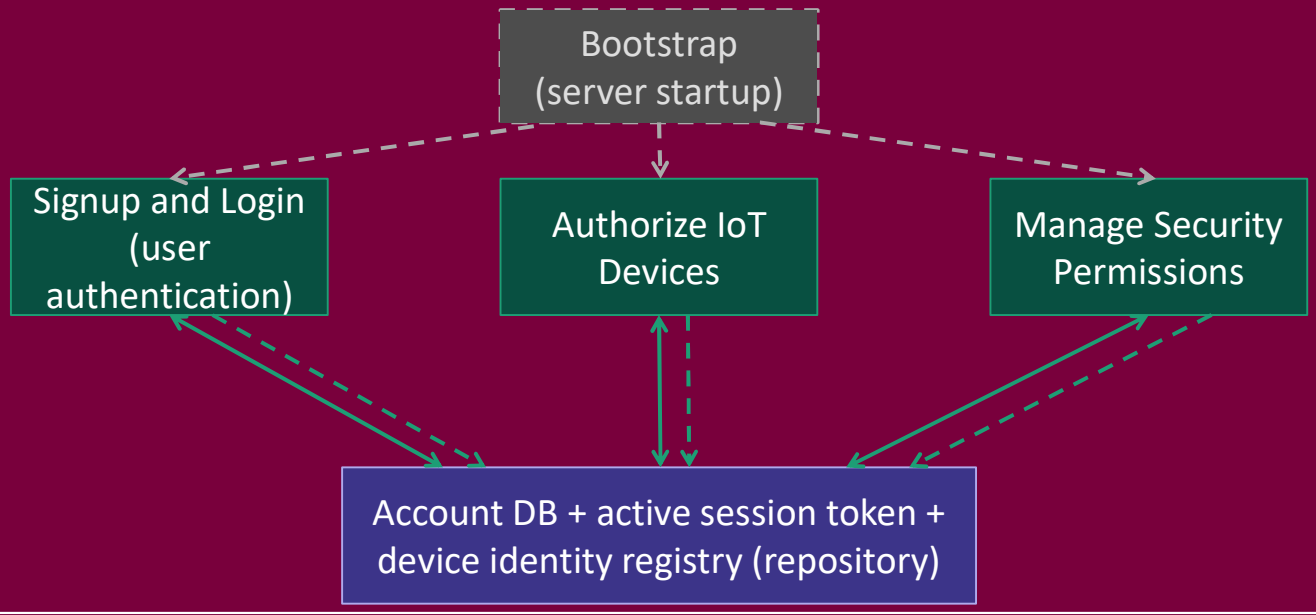
Rationale for Architecture Choices

What?	Architecture Style	Why?
Telemetry Management	Pipe and Filter	• Raw sensor reading validated for missing fields, out-of-range values, and timestamp drift before persisting • Each filter is stateless and independent



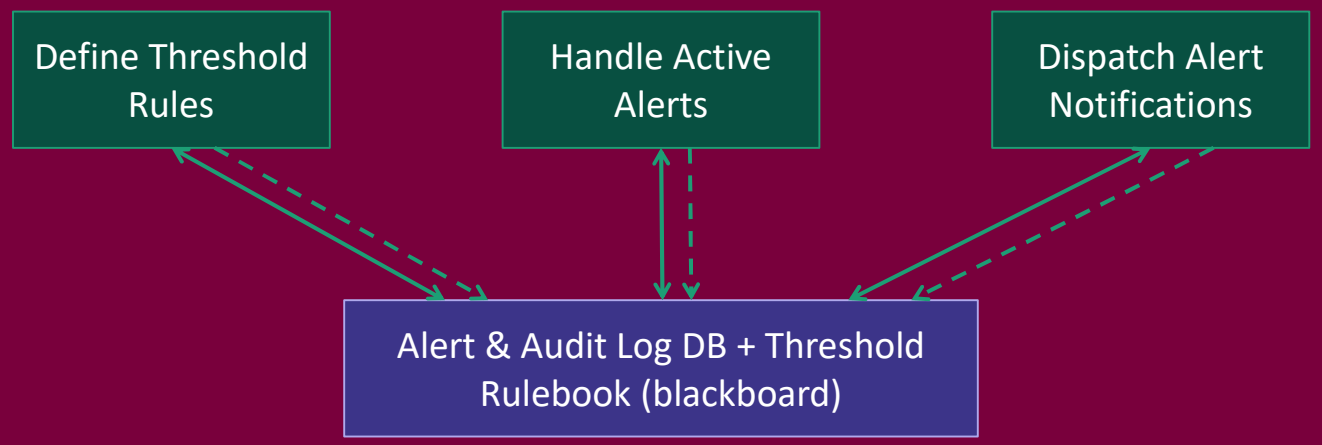
Rationale for Architecture Choices

What?	Architecture Style	Why?
Access Management	Repository	• Signup, IoT authorization, and security permissions all read and write to a shared repository deterministically

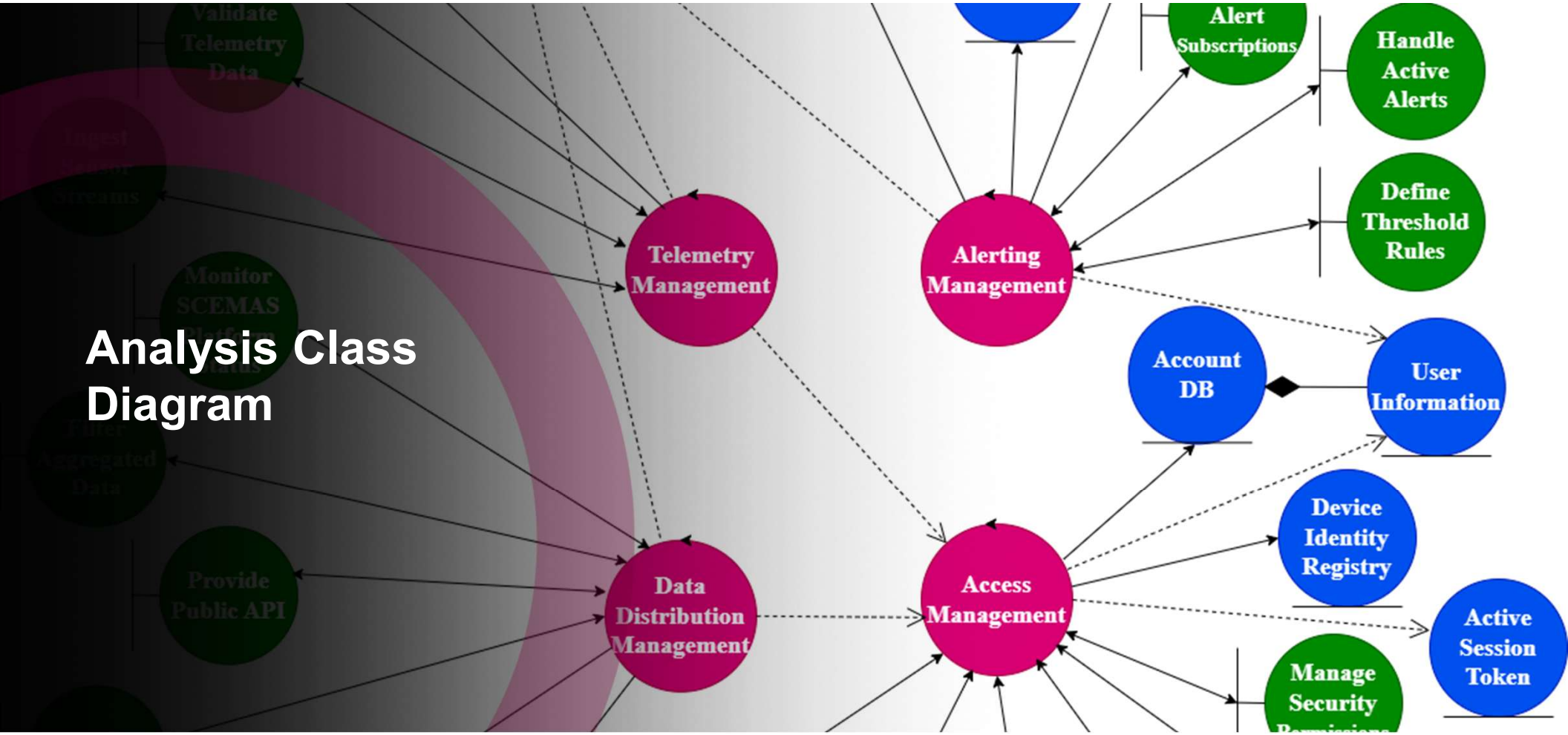


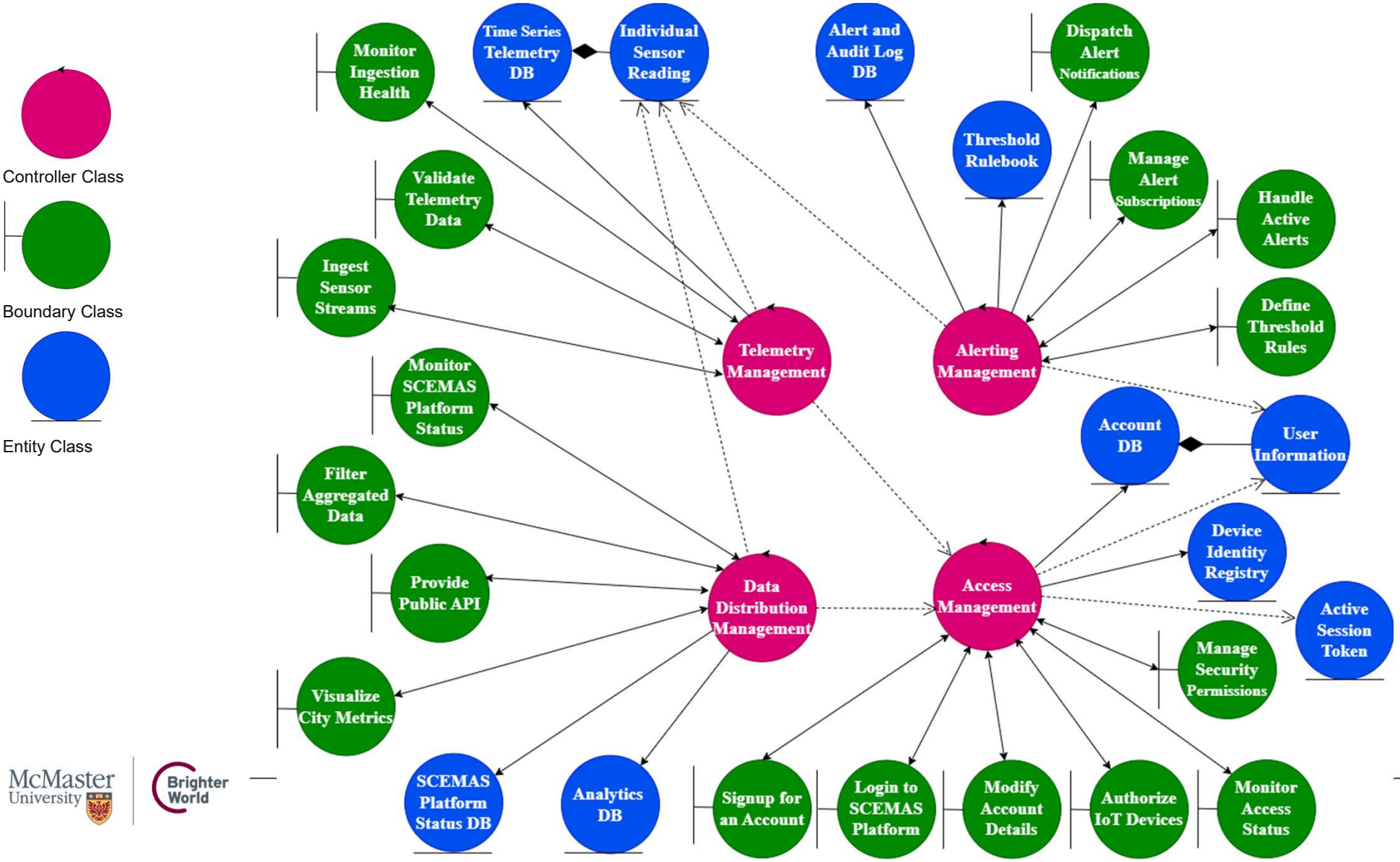
Rationale for Architecture Choices

What?	Architecture Style	Why?
Alerting Management	Blackboard	• Air quality, temperature, humidity, and noise rule evaluators **non-deterministically** read and write to a shared blackboard • Alert moves through triggered → acknowledged → resolved



Analysis Class Diagram




Controller Class
Boundary Class
Entity Class
Monitor Ingestion Health
Time Series Telemetry DB
Individual Sensor Reading
Alert and Audit Log DB
Dispatch Alert Notifications
Threshold Rulebook
Manage Alert Subscriptions
Handle Active Alerts
Validate Telemetry Data
Ingest Sensor Streams
Telemetry Management
Alerting Management
Define Threshold Rules
Monitor SCEMAS Platform Status
Account DB
User Information
Filter Aggregated Data
Device Identity Registry
Provide Public API
Data Distribution Management
Access Management
Active Session Token
Manage Security Permissions
Visualize City Metrics
SCEMAS Platform Status DB
Analytics DB
Signup for an Account
Login to SCEMAS Platform
Modify Account Details
Authorize IoT Devices
Monitor Access Status

Moving onto the live demo

During the demo, we will show how our design meets requirements